

Zero-Manual-Annotation Lane Perception for Automated Driving: Engine-Derived Ground Truth, an Explicit Counting Contract, and Pre-Registered Single-Factor Evaluation

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Abstract

Lane perception for automated driving is bottlenecked by two problems that are usually conflated: the cost of pixel-accurate manual annotation, and the absence of an agreed definition of what a "lane line" is. We present a closed-loop research programme that attacks both. First, ground truth is derived entirely from the simulator's own rendering and physics state, and admitted only through four explicit certification gates; the manual annotation budget is zero by construction. Second, we make the evaluation definition explicit and versioned: a *counting contract* gives every ratio an explicit numerator and denominator, and a *protocol* fixes the geometric scope of a line candidate (lateral extent, same-side merging, paint-appearance). We then evaluate one segmentation arm (six seeds) on two disjoint reference sets, on a road-disjoint one-shot held-out set, and in closed-loop driving. Three findings follow. (i) The two protocol definitions each fail a *different* gate on the same data, and the disagreement is definitional, not model-quality: under the paint-scope definition the adopted protocol clears all five gates on both sets (coverage 0.861, identity 0.716, precision 0.659, recall 0.885, role 0.794 on the limited-class pool), while the label-scope reading of the same checkpoints fails recall (0.559 on the development pool) because 30% of annotated line pixels are dark paint with no local contrast. (ii) The one-shot held-out confirmation does **not** reproduce those numbers (label-scope recall 0.460, precision 0.763), and we report that as the headline result of the independent test rather than hiding it. (iii) In closed-loop driving the same definition change removes all centre-line crossings (13 $\rightarrow$ 0 frames) at the cost of availability (sensor lane-source rate median 0.851 $\rightarrow$ 0.292), and the driving acceptance gate still fails 0/4 runs; we localise the remaining blocker with measurements rather than conjecture: a static deadlock in which a stationary, off-centre car is refused every path by a 4 m body-sweep rule. Throughout, changes are pre-registered, tested one factor at a time, and judged by rules fixed before the runs; negative results are reported with the same detail as positive ones. All figures and verdict files are regenerated from the recorded run artefacts.

Index Terms—Lane detection, ground-truth generation, evaluation protocol, automated driving,

I. Introduction

Lane-keeping is the lowest-level competence of an automated driving stack, and it is still surprisingly hard to evaluate. Two obstacles recur in the literature. The first is annotation cost: pixel-accurate lane labels are expensive, so public datasets are small relative to the diversity of road appearance, and every new operational domain demands a new labelling campaign. The second is definitional: papers report "lane detection" accuracy against ground truth that may include or exclude faded, occluded, dashed, yellow, or shadowed markings, and may or may not count markings beyond the ego lane. Two systems can therefore disagree by tens of accuracy points while disagreeing about nothing except the definition of the object they detect.

This paper reports a research programme built around those two obstacles. It is not a new network architecture; it is an attempt to make the *data* and the *evaluation* rigorous enough that architectural progress can be measured. The programme has three components.

Engine-derived ground truth. The perception stack is developed inside a commercial vehicle simulator whose renderer exposes both images and the semantic and geometric state that produced them. We derive lane ground truth from that state (paint material and geometry), never from human strokes, and we admit a frame only if it passes four explicit certification gates: a credential gate (the label source must be declared and verifiable), a paint-appearance gate (the pixels claimed as line must look like paint in the image), a road-type gate (the frame must carry the road-type column that makes pavement decidable), and a negative-evidence gate (frames claimed to contain no line must show evidence of their own, not merely the absence of a label). The manual annotation budget of the entire programme is zero by construction; this is a design constraint, not an achievement, and Section VII is explicit about what it costs us.

An explicit counting contract. Every reported ratio is defined by a counting contract that names its numerator and denominator over an explicit set chain: frames with candidates, extracted candidate instances, reference instances present in the annotation, matched instances, instances whose lateral placement agrees, and instances whose left/right role agrees. Coverage, identity and role are therefore instance-level ratios

with stated denominators; precision and recall remain pixel-level and are reported in two *scopes* (label pixels, and paint-like pixels) because the two answer different questions.

Versioned protocol definitions with hash-gated evaluation.

A protocol fixes the geometric scope of a line candidate and is a first-class, versioned artefact. A protocol change alters the evaluation hash, and both the frozen held-out set and the one-shot confirmation record that hash, so a confirmation produced under one definition cannot be silently attributed to another. This mechanism caught a real defect during the work: before it was fixed, the hash did not follow the protocol, and a set sealed under the old definition would have authorised a confirmation under the new one.

Method discipline. Every change in this paper was pre-registered with its reading rule before the runs; the rules include safety non-inferiority read as a maximum over runs, an explicit refusal to treat an unmeasured quantity as a pass, and a prohibition on relaxing the frozen gates. Changes were applied one factor at a time and judged against alternating control/factor runs. Where a pre-registered rule produced a verdict we considered too coarse, we report the verdict as given and record the rule's defect rather than reinterpreting the result; one such defect — safety metrics that are frame *counts* and therefore scale with how far the car travelled — is analysed in Section V-F and quantified in 18.

Contributions. (1) A zero-manual-annotation, gate-certified ground-truth pipeline for lane perception, with its costs stated. (2) The counting contract, and the demonstration that the two natural readings of "line" fail different gates on the same data — a definitional disagreement worth 30% of annotated pixels. (3) A protocol/version/hash mechanism that binds a held-out set to the definition under which it is read. (4) A one-shot, road-disjoint confirmation with an explicit UNKNOWN policy, whose negative result (no reproduction of the development-set numbers) we report as the finding. (5) A measured localisation of why a perception stack that passes its offline gates still fails a closed-loop acceptance gate, including a static-deadlock mechanism and the observation that the "on-pavement" gate was in fact being driven by a neural obstacle layer rather than by pavement evidence.

II. Related Work

Lane detection methods. Modern lane detection is dominated by deep segmentation and by parametric or anchor-based curve regression. Encoder-decoder segmentation with dilated convolutions and a discriminative loss (LaneNet [1]) and message-passing across rows (SCNN [2]) established the segmentation line of work that the present system follows; cross-layer refinement with anchor-based heads (CLRNet [3]) improved curve accuracy and speed. Chinese-language work in the same family explores lightweight variants of CLRNet [4],

B-spline end-to-end regression [5], graph/transformer hybrids [6], transformer with cross-attention [7], parallel detection over image sequences [8], multi-scale feature aggregation [9], DeepLabV3+ variants [10], symmetry-prior and key-point methods for complex scenes [11], and 3-D lane estimation with convolutional architectures [12]. These works share our object of study but differ in what they optimise: they report accuracy against datasets whose annotation conventions differ from each other, which is exactly the ambiguity this paper makes explicit.

Datasets and ground truth. TuSimple [13] and CULane [2] remain the standard benchmarks; their annotation conventions (lane presence per row, presence of far-field markings, treatment of occlusion) differ, and neither reports the annotation's *appearance* properties. Ground-truth generation has been studied as a problem in its own right: Borkar et al. [14] built an automated ground-truth process around time-slicing visualisation for a night-time lane detector, and McCall and Trivedi [15] surveyed and evaluated video-based lane estimation with an explicit eye on evaluation methodology. Our work continues that line with two additions: truth derived from engine state rather than from imagery, and a counting contract that makes the denominators of every reported ratio inspectable.

Automatic labelling and simulation. Self-training and auto-labelling are widely used to reduce annotation cost, but their evaluation inherits the definitional problem above. Simulator-derived labels are common in synthetic benchmarks; less common is the discipline of certifying each generated label against image evidence, which we adopt here because a label that cannot be recovered from the image is not a learnable target.

Evaluation discipline. Pre-registration is standard in empirical sciences and rare in perception research. Our contribution here is not the concept but a worked, fully recorded instance: a pre-registration, an alternating single-factor A/B, a fixed reading rule, and a verdict per factor — including five factors whose verdict was negative.

III. System and Method

A. Overall architecture

The stack is a perception-led driving system. A camera ring (eight mounts) and a LiDAR range image feed a semantic segmentation head whose line output is converted to world-space markings; a lane reference is selected from those markings under the protocol definition; a planner produces a path; a safety monitor arbitrates the tick against hard checks (pose, boundaries, occupancy, obstacle risk) and publishes a speed and level; the controllers consume the arbitrated path. The iron rule of the stack is that the lateral reference comes from perception only — the map answers "where to", never

"where the lane is". 1 shows the candidate-scope pipeline with the measured revocation rate discussed in Section V-E.

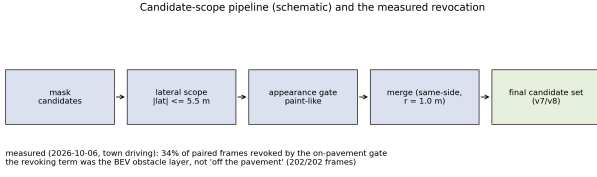


Fig. 1. Candidate-scope pipeline (schematic) with the measured revocation rate.

B. Engine-derived ground truth and its certification gates

Ground truth is produced by exporting, for each collected frame, the engine's own line geometry projected into the camera, together with a credential that names the label source and a content hash. A frame is admitted only when: (i) the credential is present and parseable, and the label source is declared (frames whose provenance cannot be named are excluded rather than guessed); (ii) the pixels claimed as line satisfy a paint-appearance predicate — white paint requires high value, low chroma and a red/blue balance; yellow paint requires a hue ratio band; the predicate is a single-source module so that a threshold change is a versioned change, not a local edit; (iii) the frame carries the road-type column, without which "on pavement" is undecidable; (iv) a frame claimed to contain no line carries positive evidence of its own (a certified negative), so that "no label" is never read as "no line". Frames failing any gate are excluded from both training and evaluation and are counted in the audit, not silently dropped.

C. The counting contract

Let P be the set of frames that carry at least one candidate, C the extracted candidate instances, C_{out} the candidates that fall on frames outside P , $R \subseteq C$ the reference instances available from annotation, $M \subseteq R$ the matched instances, $L \subseteq M$ those whose lateral placement agrees, and $A \subseteq L$ those whose left/right role agrees. The reported instance-level metrics are coverage $|R|/|C|$, identity $|M|/|R|$ and role agreement $|A|/|L|$; pixel-level precision and recall are reported in two scopes (label, paint). 2 shows the contract. Two properties matter for reproducibility: every ratio has an explicit denominator, and the frame-level set P is reported separately from the candidate-level sets, so a run with few candidate-bearing frames cannot masquerade as a good one.

Counting contract v5 (schematic): every ratio has an explicit numerator and denominator

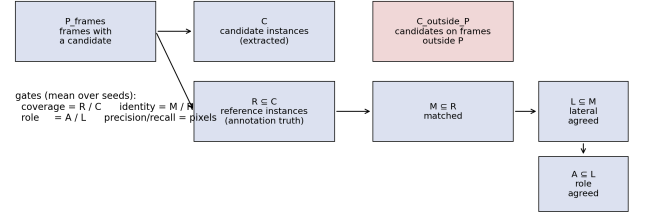


Fig. 2. Counting contract (schematic): every ratio has an explicit numerator and denominator.

D. Protocol definitions

A protocol fixes three things about a line candidate: its **lateral scope** (candidates farther than a threshold from the ego's own lane are not "my lane's lines"), whether **same-side near-duplicate markings are merged** into one instance, and whether an **appearance gate** removes candidates whose pixels do not look like paint. The frozen baseline protocol applies none of the three (lateral scope off, merging off, appearance off); the adopted protocol applies all three with a lateral threshold of 5.5 m, a merge radius of 1.0 m and the appearance predicate of Section III-B. 3 shows the switch matrix and the live hash; the hash is a function of the active protocol, so evaluation artefacts cannot be reattributed across definitions.

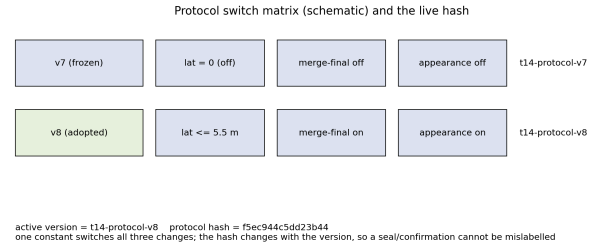


Fig. 3. Protocol switch matrix (schematic) and the live evaluation hash.

E. Training-composition design space

Because ground truth is generated rather than annotated, the composition of the training pool is a first-class experimental variable. We study it along one axis: the **negative dose**, i.e. how many certified line-free frames are mixed into the pool relative to the base composition, together with composition variants that place markings in specific geometric relations (near/far pairs, same-side pairs). Section V-C reports the resulting dose-response.

F. Held-out final set: sealing, hash gating, one-shot consumption

An independent confirmation set is composed under three constraints: it must be **road-disjoint** from every training pool (identity taken from the generation metadata, not from directory names), it must be certified by the gates of Section III-B, and its content must be frozen by a digest over per-frame

hashes. Sealing records the frame digests, the protocol hash and the composition; access is then gated on the protocol hash, and a single confirmation **consumes** the set: after a successful confirmation the set may not be read again, and a refused access is recorded in a ledger without consuming it. The confirmation program evaluates the candidate on the sealed frames and records which quantities were measured; quantities that were not measured are recorded as UNKNOWN and may not be reported as confirmed.

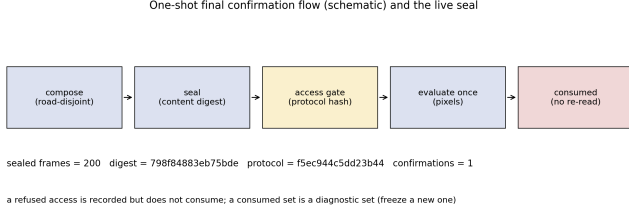


Fig. 4. One-shot final-confirmation flow (schematic) with the live seal.

G. Driving stack and safety arbitration

In closed loop the same perception feeds a planner and a safety monitor whose arbitration ladder evaluates rules in a fixed order; soft rules accumulate speed caps, while a rule whose worst level is `minimal_risk` stops the tick. 5 lists the ladder. Two rules matter for Section V-E: the **planned body-sweep rule**, which stops the tick when the body sweep of the planned path would cross a lane boundary closer than a hard threshold (4 m), and the **on-pavement gate**, which revokes the perception lane when the lane centre does not lie on observed drivable surface.

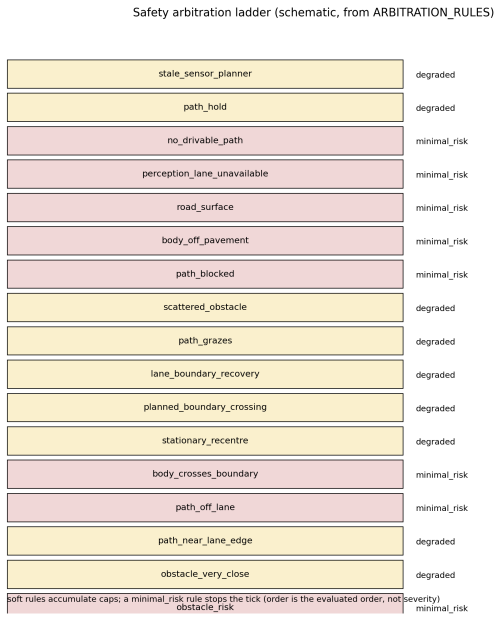


Fig. 5. Safety arbitration ladder (schematic, from the monitor's rule registry).

IV. Experimental Setup

Reference sets. Two disjoint sets are used for the primary acceptance: a development pool of 9 certified scenes (117 frames) and a limited-class pool of certified scenes chosen for near-field background structure (24 frames per arm). Neither shares frames with training pools. A third, road-disjoint set of 200 frames (96 line-bearing) is sealed for the one-shot confirmation of Section V-D.

Arms. All segmentation arms share one initialisation, one budget and one recipe; they differ only in training-pool composition (the dose arms) or in a single post-processing factor (protocol components). Six seeds are used for acceptance; composition arms are reported with their seed counts.

Metrics and gates. The frozen gates are coverage ≥ 0.80 , identity ≥ 0.60 , precision ≥ 0.40 , recall ≥ 0.70 and role agreement ≥ 0.70 ; recall is read in the paint scope for the adopted protocol and in the label scope for the baseline, and both readings are always reported. Gate verdicts use the mean over seeds with per-seed reporting, which is the repository's existing rule; a per-seed failure is reported even when the mean passes.

Driving experiments. Closed-loop runs use one fixed scenario (a town link), alternating control and factor arms, four runs per arm; hard targets are the benchmark's fixed checks (frames, reversing, centre and edge crossings by reference line *and* by body sweep, stall, on-road, collisions, goal). Reading rules were fixed before each round: hard gates require 4/4, safety items are compared as the maximum over runs, availability as median with range, and an UNKNOWN never releases a gate.

Discipline. Every factor in Section V was pre-registered with its reading rule; no threshold was relaxed at any point; every verdict file, manifest and per-frame telemetry trace is retained, and every figure in this paper is regenerated from those files by two scripts.

V. Results

A. Acceptance under two definitions

Table I reports the two definitions on both sets. Under the baseline definition the identity gate fails on **both** sets (0.433 dev, 0.414 limited-class) and the limited-class set additionally fails precision (0.402) and role (0.674); under the adopted definition all five gates pass on both sets (coverage 0.956/0.861, identity 0.665/0.716, precision 0.894/0.659, paint recall 0.812/0.885, role 0.883/0.794). 6 shows the same numbers against the thresholds.

Set	Definition	Coverage	Identity	Precision	Recall	Role
Dev (9 scenes, 117 frames)	baseline (label scope)	0.952	0.433	0.610	0.736	0.868
Dev	adopted (paint scope)	0.956	0.665	0.894	0.812	0.883
Limited-class (24 frames)	baseline (label scope)	0.819	0.414	0.402	0.804	0.674
Limited-class	adopted (paint scope)	0.861	0.716	0.659	0.885	0.794

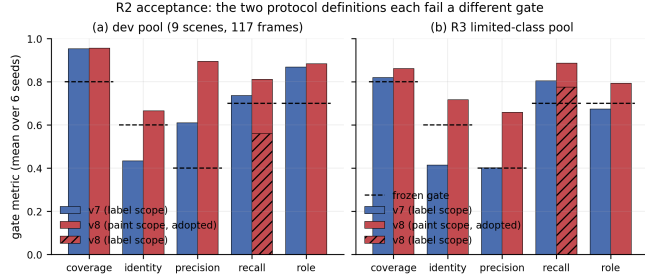


Fig. 6. Gate metrics against frozen thresholds for the two protocol definitions, development pool (a) and limited-class pool (b).

B. The disagreement is definitional

The two definitions differ by one gate each, and the difference is traceable to pixels, not to model quality. 7 shows, per seed, the label-scope and paint-scope recall of the *same* checkpoints, with the fraction of label line pixels that are not paint overlaid: that fraction is ~ 0.30 , i.e. roughly a third of annotated line pixels are dark paint (shadowed asphalt paint) that carries no local contrast, and those pixels are exactly the ones the paint-scope definition excludes and the label-scope definition demands. The same checkpoints therefore read as 0.559 or 0.812 recall depending on the definition — a 25-point swing with no change to the model.

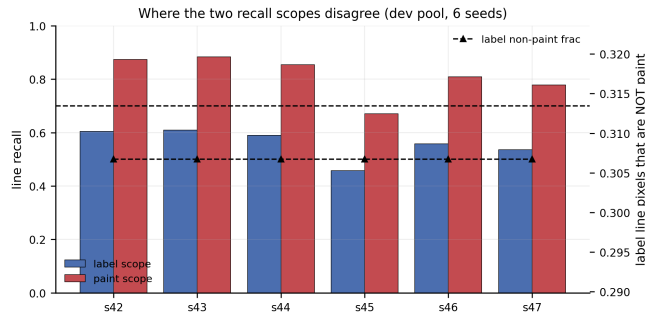


Fig. 7. Per-seed label-scope versus paint-scope recall with the non-paint fraction of label line pixels (development pool).

Two further single-factor results constrain how the adopted definition behaves. 8 (boundary map) sweeps mask dilation: dilation buys recall in both sets but costs precision, and the limited-class set pays about three times more (precision 0.65 $\rightarrow$ 0.39 at one pixel of dilation, against 0.89 $\rightarrow$ 0.84 on the development pool), so no single dilation value clears both sets. 9 shows the appearance gate acting exactly as designed: precision rises sharply (dev 0.650 $\rightarrow$ 0.884; limited-class 0.387

$\rightarrow$ 0.650) and IoU improves on the limited-class set (0.347 $\rightarrow$ 0.532) at the cost of recall (dev 0.766 $\rightarrow$ 0.595).

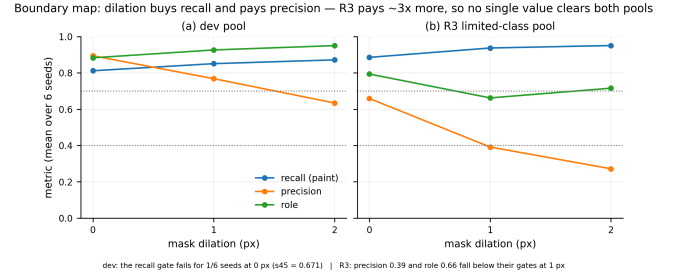


Fig. 8. Boundary map: dilation buys recall and pays precision; the limited-class set pays about three times more.

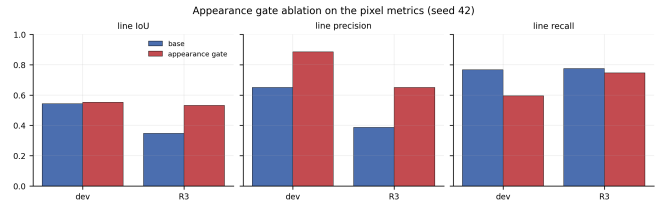


Fig. 9. Appearance-gate ablation on the pixel metrics (line IoU, precision, recall) for both sets, seed 42.

C. Training-composition dose response

10 shows the identity rate against the negative dose, per seed and as a mean: 0.707 (base composition, one seed), 0.711 (4 $\times$, six seeds, 0.667–0.732), **0.763 (6 $\times$, one seed)** and 0.701 (8.5 $\times$, six seeds, 0.655–0.744). The response is non-monotonic with a peak at 6 $\times$ and a return to baseline at 8.5 $\times$; the 6 $\times$ arm is the delivered candidate's composition. 11 sweeps the lateral scope instead: tightening it from 6.0 m to 3.0 m raises identity (dev 0.582 $\rightarrow$ 0.713; limited-class 0.581 $\rightarrow$ 0.758) and removes candidates (477 $\rightarrow$ 166 and 166 $\rightarrow$ 127), while role agreement is flat — the lateral scope is an identity lever, not a role lever. Two further single factors were rejected: stochastic weight averaging over the last five epochs rescued no seed (12) and a Tversky loss-weight arm tripled the false-positive frame rate on the negative side (11.5% $\rightarrow$ 51.9% of eligible frames; 329 $\rightarrow$ 1730 false-positive pixels; maximum component 320 $\rightarrow$ 808 px), which is why the delivered arm keeps the balanced loss.

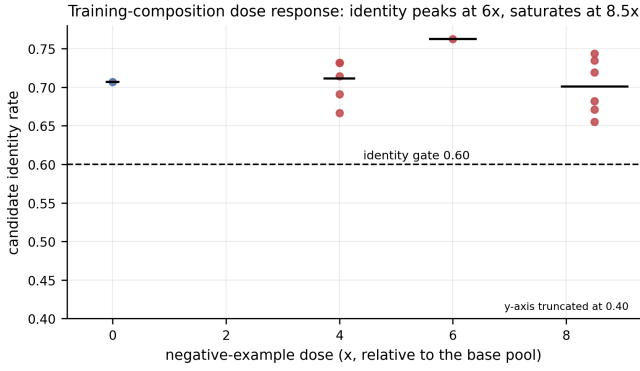


Fig. 10. Dose response of the candidate identity rate (per seed and mean).

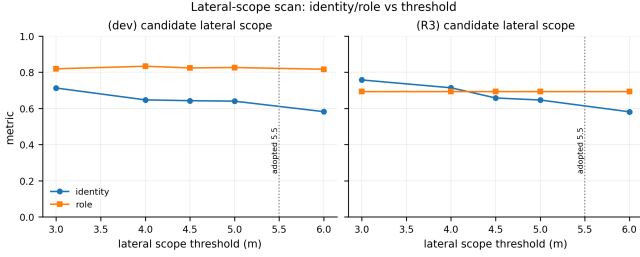


Fig. 11. Lateral-scope scan on both sets: identity and role versus threshold, with the number of retained candidates.



Fig. 12. Rejected single factor: last-five-epoch weight averaging rescues no seed (paint-scope recall).

D. One-shot held-out confirmation

The sealed set was read once, by the delivered candidate. Overall label-scope recall is **0.460** and precision **0.763** (IoU 0.403; road IoU 0.633); per group, three of seven groups sit at recall 0.17–0.20 while precision stays 0.86–0.92 in the same groups, one group carries no line ground truth (recall undefined, recorded as such), and the remaining groups range 0.43–0.63. On the negative side 22 of 105 eligible frames carry false line pixels (20.9%), but those pixels are 0.084% of eligible pixels, the largest connected component is 1164 px, and no false candidate appears inside the control region. 13 shows the confirmation. Two properties of this result deserve emphasis. First, it does **not** reproduce the acceptance numbers of Section V-A: the independent set is harder and its group composition differs (104 of 200 frames come from certified line-free packages), so the delivered candidate's claim is bounded by the numbers reported here rather than by the acceptance table. Second, the confirmation records that the

instance-level probes (identity, role, reference coverage) were **not** run on this set; they are UNKNOWN and are not reported as confirmed.

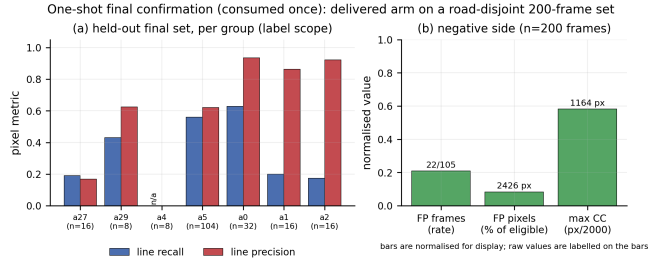


Fig. 13. One-shot final confirmation on the sealed road-disjoint set: per-group metrics (a) and negative-side diagnostics (b).

E. Closed loop

Definition change in closed loop. With the baseline definition the closed-loop runs show centre crossings of 8, 8, 8 and 13 frames; with the adopted definition the same scenario shows 0, 0, 0 and 0. The adopted definition pays for this with availability: the sensor lane-source rate falls from a median of 0.851 (range 0.231–0.975) to 0.292 (0.097–0.524), and the paired-lane rate from 0.890 (0.809–0.986) to 0.528 (0.130–0.995). 14 shows the trade-off. Both effects are consistent with the offline finding that the baseline keeps far and non-paint candidates whose pairings pull the car across the lane centre.



Fig. 14. Closed-loop trade-off: centre crossings versus sensor lane-source rate, four runs per arm.

Acceptance failure and its mechanism. The driving acceptance gate (goal + perception-led lateral mode + strict arbitration, four runs) fails **0/4**: stall fraction 0.93–1.00, travelled 0.9–14.5 m in 120 s, 76.5–88.8 m remaining to goal; the safety-side checks that were measured held (0 collisions, 0 off-road frames, 0 reversing). 15 shows the per-run hard-target matrix over the recent runs. We then localised the blocker by measurement rather than conjecture. Over 12 town runs, the frames whose reason is "planned vehicle body crosses lane boundary" number 512; in those frames the car is stationary in 96% of cases, the current body is inside the lane in 512/512, and the planned crossing sits at a median of 2.50 m (509/512 within 2–4 m) — i.e. just inside the 4 m hard-stop threshold.

The car is parked 0.86 m off the lane centre with its heading $\sim 11^\circ$ off the lane bearing, so the inflated body sweep of *any* forward path crosses the boundary within ~ 2.5 m: every path is refused, and the car can never re-centre. 16 shows the anatomy.



Fig. 15. Driving hard-target checklist over the most recent town runs (green = pass, red = fail; an unmeasured target counts as fail).

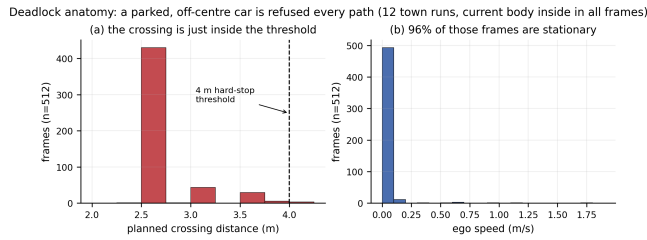


Fig. 16. Deadlock anatomy: the planned crossing sits just inside the 4 m threshold (a) and the car is stationary in 96% of those frames (b).

Factor	Mechanism engaged?	Travelled (median)	Stall (median)	Worst-case body-cross frames (control $\rightarrow$ factor)	Verdict
Bounded hold window	yes (reuse 7 $\rightarrow$ 42 frames)	3.15 $\rightarrow$ 8.5 m	0.986 $\rightarrow$ 0.958	29 $\rightarrow$ 30	rejected
Path re-centring	yes (39 frames)	5.1 $\rightarrow$ 5.75 m	0.984 $\rightarrow$ 0.980	84 $\rightarrow$ 39	rejected (one item worse)
On-pavement gate (obstacle term removed)	yes (availability 0.07–0.52 $\rightarrow$ 0.36–0.90)	6.85 $\rightarrow$ 14.1 m	0.966 $\rightarrow$ 0.950	23 $\rightarrow$ 66	rejected
Delivered perception arm	no (no separation)	—	0.979 $\rightarrow$ 0.995	36 $\rightarrow$ 49	rejected
Pavement re-centring (as proposed)	not implemented	—	—	—	refuted by prior measurement

Why the lane was being dropped. The on-pavement gate revokes the perception lane when the lane centre does not lie on observed drivable surface. Instrumenting the gate's evidence over two diagnostic runs (415 settled frames) shows that of 202 revoked frames, **all 202** were revoked by the obstacle term (median 6 obstacle cells of ~ 12 observed samples), while the number of samples that were observed but outside the drivable mask is zero in 192 of them. The obstacle layer is populated from a neural bird's-eye-view head, so on this link the "on-pavement" gate was in effect an obstacle-prediction veto on the lane centre. 17 shows the evidence and the availability funnel: in the baseline configuration only 116 of 415 settled frames (28%) have a perception lane and only 79 (19%) a planner path.

Five single factors, all negative. We pre-registered and ran five driving-layer factors against this blocker (Table II). A bounded "hold" window that lets a verified trajectory be reused past its offer age raised motion (reuse 7 $\rightarrow$ 42 frames; travelled 3.15 $\rightarrow$ 8.5 m) but worsened the worst-case body-crossing count (29 $\rightarrow$ 30). Pulling the planned path onto the perceived lane centre engaged in 39 frames and improved the worst case substantially (centre crossings 3 $\rightarrow$ 0; body crossings 84 $\rightarrow$ 39) but worsened one right-side item (0 $\rightarrow$ 2). Removing the obstacle term from the on-pavement gate — see below — doubled travelled distance (6.85 $\rightarrow$ 14.1 m) and halved the distance to goal (83.5 $\rightarrow$ 76.6 m) but worsened body crossings (23 $\rightarrow$ 66) while the car moved twice as far. Swapping in the delivered perception arm produced no separation on path availability (0.179 $\rightarrow$ 0.200, overlapping ranges) and a worse worst case (36 $\rightarrow$ 49). All five verdicts follow the pre-registered maximum rule and are reported as negative; none of the factors was adopted.

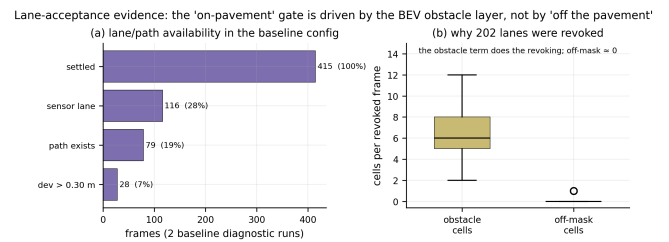


Fig. 17. Lane-acceptance evidence: baseline availability funnel (a) and the evidence behind the revoked lanes (b).

F. Real-time behaviour and a metric defect

Control cadence. Per-frame telemetry shows the perception-and-planning tick costs 377–432 ms, of which 355 ms is sensor reading (camera ring 139–162 ms, range image 216–239 ms) and 11–13 ms is planning; the neural heads cost 3–5 ms. The renderer itself runs at ~ 50 fps (20 ms per frame), so the simulation is not the bottleneck: the control loop is paced at 2

Hz by design and is in any case tick-bound at ~ 2.5 Hz. Between ticks a sub-step path re-runs the controllers at up to 15 Hz, but it is gated on a path having been chosen: in the stall regime (no path) no sub-steps run and commands are issued every ~ 0.5 s, while in an arm where paths exist 12 sub-steps

per frame yield an effective ~ 31 Hz. Table III summarises. The practical consequence is that the perceived "stutter" of the vehicle is a control-cadence artefact of the path-availability blocker, not a rendering or compute failure.

Run	Tick (ms)	Sensor read (ms)	Renderer frame (ms)	Sub-steps per frame	Effective command rate
Baseline acceptance (no path)	432	401	20	0	~ 2.2 Hz
Lane-gate factor arm	377	355	20	1 (max 5)	~ 5 Hz
Path-re-centring arm	403	~ 370	20	12	~ 31 Hz

A defect in the safety reading rule. The safety item used to judge every driving factor is a frame *count* of body-centre crossings, and a factor arm can travel twice as far as its control. Fig. 17 quantifies the consequence: the same arm that halves the per-metre crossing rate can show a larger raw count. This is why, after the first two verdicts, we re-registered a per-distance reading (crossings per 100 m) alongside the raw maximum; the earlier verdicts stand as given, and we flag the rule defect as a methodological result rather than editing them.

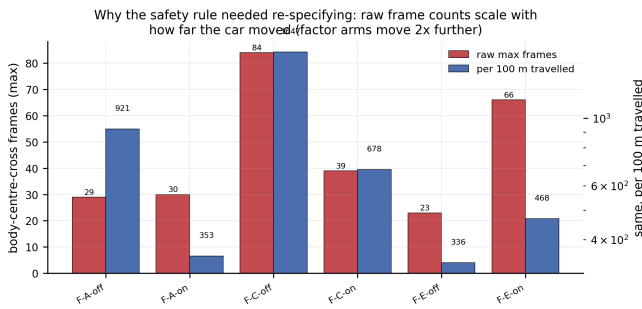


Fig. 18. Raw crossing counts versus the same quantity normalised per 100 m travelled (log scale).

VI. Discussion

What generalises. Four artefacts from this programme seem reusable beyond the specific stack. (i) **The counting contract:** reporting every ratio with its denominator, and keeping frame-level and instance-level sets separate, removed an entire class of "good number, empty pool" errors and made per-scene failures visible instead of averaged away. (ii) **Protocol versioning with a hash bound to the sealed set:** the mechanism is cheap and it caught a real mislabelling defect; without it, an evaluation produced under one definition would have been attributed to another. (iii) **Appearance certification of generated labels:** requiring that a claimed line be recoverable from the image turned "annotation noise" from a tuning problem into an admission decision. (iv) **Pre-registered single-factor A/Bs with a fixed reading rule,** including an explicit refusal to treat unmeasured quantities as passes; this is what allowed five negative results to be reported as results rather than as failures.

What did not work. The definitional gap is not closable by engineering: the 30% of annotated line pixels that are dark paint are invisible to any local-contrast model, and training on them teaches hallucination — a documented negative result in our logs. The offline gates do not transfer to closed-loop acceptance automatically: the same perception that clears every offline gate leaves the car unable to move, for reasons (a static body-sweep deadlock) that no offline metric expresses. Finally, availability and safety trade against each other along the definition axis in a way that a single scalar cannot capture: the adopted definition removes all centre crossings and halves availability.

Implications for evaluation practice. Three practices would have changed our results materially. Report the *definition* alongside the metric, and make the definition a versioned artefact. Report per-distance or per-opportunity normalisations for safety counts, because motion confounds raw counts. And run one independent, one-shot confirmation whose composition differs from the development pool; our confirmation's negative result is the single most informative measurement in the programme.

VII. Limitations and Threats to Validity

No human ground truth. By design the programme uses zero manual annotation. The cost is that we cannot report agreement with human labels, and the definitional claims rest on our own certified truth and on the appearance analysis of Section V-B rather than on an adjudicated reference. A human-labelled validation subset would strengthen the definitional argument and is left to future work.

Single simulator, single map link. All perception data come from one simulator and, for the driving results, one town link. The definitional gap and the deadlock mechanism are measured on that link; their magnitude elsewhere is unknown. The offline sets span several certified scenes but not several maps.

Six seeds, and unequal seed counts. Acceptance uses six seeds; composition arms are reported with their seed counts (one to six), and the $6\times$ dose point rests on a single seed. Dose conclusions are therefore about a peak location, not about a precise effect size.

The held-out set is consumed. The one-shot confirmation was performed once, as designed, and the set may not be read again; any future claim requires a newly sealed set. The instance-level probes were not run on it and remain UNKNOWN.

Artefact retention. Per-epoch training snapshots and superseded run directories were removed in a storage pass; every verdict file, manifest, per-frame telemetry trace, sealed frame and delivered checkpoint is retained, so all results and figures in this paper are reproducible from the retained artefacts, but some intermediate training states are not.

Driving results are from a research stack. The closed-loop results describe our stack, not automated driving in general; the acceptance gate is a project-internal target, and its thresholds were fixed before the runs and never relaxed.

VIII. Conclusion

We set out to make lane perception for automated driving measurable without manual annotation, and to make its evaluation honest about definitions. Engine-derived truth with four certification gates removed the annotation budget from the programme; a counting contract gave every reported ratio an explicit denominator; and a versioned, hash-bound protocol made the definition of "line" an artefact rather than an assumption. The resulting measurements are informative precisely where they are uncomfortable: the two natural definitions of a line each fail a different gate on the same data, and the disagreement is a third of the annotated pixels; the adopted definition passes all offline gates and removes every centre crossing in closed loop, at half the availability; and the one-shot independent confirmation does not reproduce the development numbers, which we report as the headline of that test. The closed-loop acceptance gate still fails, and we localised the blocker to a measurable deadlock mechanism rather than a tuning shortfall. Every factor, including five negative ones, was pre-registered and judged by rules fixed in advance; we consider that discipline, and the recorded negative results it produced, the main methodological contribution of this work.

Reproducibility and Data Availability

All results are regenerated from recorded artefacts: per-run scorecards, per-frame telemetry traces, run manifests, gate verdict files, and sealed-set records. Two scripts regenerate every figure in this paper and the supplement from those files; the protocol hash, the sealed-set digest and the one-shot confirmation record are printed inside the figures so that a reader can verify which definition and which data produced them. Training pools, checkpoints, the sealed set and the audit

of removed artefacts are retained in the project repository; per-epoch snapshots removed during a storage pass are listed in a machine-readable audit file.

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